LENOVO HACKATHON · SOFTWARE · CYBERSECURITY

PHISHEYE

Scan the link before the fake page loads.

Explainable phishing and lookalike-domain detection.
One FastAPI engine. A React console. A Chrome scan-before-open gate.

Working software prototype · No hardware · No trained CNN · SQLite live

A padlock does not mean it is your bank.

ATTACKER
Fake KYC / OTP lure

→

TRUST
Fear, urgency, brand

→

FAKE LINK
HTTPS lookalike

→

VICTIM
Enters OTP / UPI

→

LOSS
Account or money

What the victim sees

SMS or WhatsApp: “Your SBI account will be blocked. Complete KYC in 10 minutes. Share OTP.”

The page looks official. HTTPS is present. The warning in the browser is technical. The money still leaves.

Who we designed for

First-time UPI users and people who will not parse an SSL mismatch.

Simple View is written in OTP language, not certificate jargon.

Feeds are late. Certificates are not brands. Panic is a signal.

Technical signals

Lookalike host, new domain, TLS facts, SSRF-safe unwrap. Absence of a feed is not a clean bill of health.

Social engineering

Urgency, fear, OTP pressure — English + Hinglish phrase rules, not a language model.

Threat intelligence

A confirmed hit floors the score. Heuristics cannot talk it down. A miss is not SAFE.

Explainable output

Simple View for humans who will not read a certificate. Technical View for operators.

In-browser act

Scan first. LOW opens. MEDIUM asks. HIGH / CRITICAL stay closed.

Honesty bound

No competitor ranking. No published F1. No trained CNN in this repository.

PHISHEYE scores the lure. Then it tells the user what to do.

INPUT
URL · paste · QR · screenshot

→

ANALYZE
Normalize + signals

→

FUSE / FLOOR
Weights or intel floors

→

RESULT
LOW → CRITICAL

→

ACT
Explain · store · gate

Protects

People who will click a KYC SMS.
Operators on the React console.
Anyone using the Chrome gate.

Analyzes (implemented)

Direct URLs
Pasted SMS / WhatsApp / email text
QR images (OpenCV, no visit)
Screenshots (QR; OCR if Tesseract exists)
Visible page links via extension

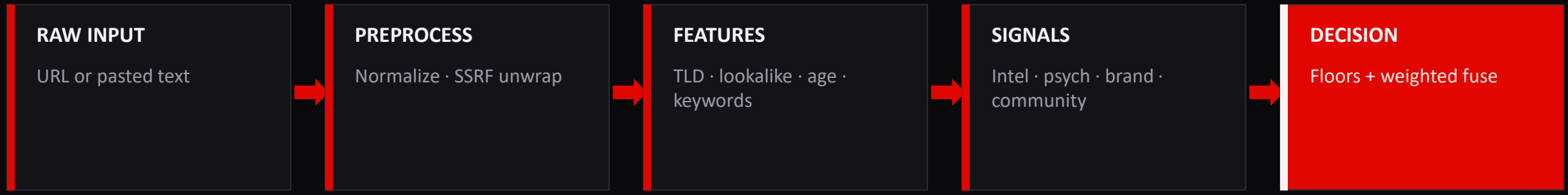
Produces

Risk 0–100 and a level
Simple View warning
Technical evidence
SQLite history
Extension allow / hold / block

Clients → FastAPI → analyzers → decide → SQLite → explain.

USER	Paste a URL or message, or navigate in Chrome.	Two clients, one brain.
API	FastAPI + Uvicorn on port 8000. Not Flask.	JWT on mutating routes. GET /health.
SERVICES	URL, brand, TLS, SSRF unwrap, QR (no visit).	Intel feeds + psychology rules.
DECIDE	URL path: intel floors cannot be averaged away.	Content path: documented fusion weights.
STORE	Live database is sqlite3 threats.db.	Simple View + Technical View out.

Explainable scoring. No trained model file in this repository.



What exists

Heuristic URL features (TLD, lookalike, age, keywords)
Phrase matching for panic / OTP language
Documented fusion weights — starting defaults, not proven-optimal
Fusion key “ml” copies the URL score so a low ML number cannot prove safety
NumPy reshapes images for OpenCV — not neural nets

What does not exist

No sklearn / TensorFlow / CNN / BERT
No labelled production dataset
No published accuracy, precision, recall, or F1
model_confidence = how many signals ≥ 20 — not accuracy
Supervised model = FUTURE after curated reports

Rules that read panic. Not a neural language model.

Example (implemented patterns)

“Your account will be blocked. Verify immediately.”

Hits: fear_tactics · urgency_detected · credential_pressure
English + Hinglish: abhi click, band ho jayega, enter otp, complete kyc

Guardrail

Psychology cannot take a site to HIGH/CRITICAL alone.
If no strong URL / domain / intel / brand / community signal, fuse caps the score at 35 (MEDIUM).

Urgency

act now · jaldi karo

Fear

account blocked

Impersonation

RBI · cyber cell

Coercion

mandatory KYC

Money pressure

pay immediately

OTP / PIN

enter otp · share otp

Countdown

within 10 minutes

Reward bait

you have won

Secrecy

do not tell anyone

Emotional ask

help me urgently

Scan first. Then maybe open.

URL

check-url

Pasted text

SMS / WA / email paste

QR

OpenCV · no visit

Screenshot

QR + optional OCR

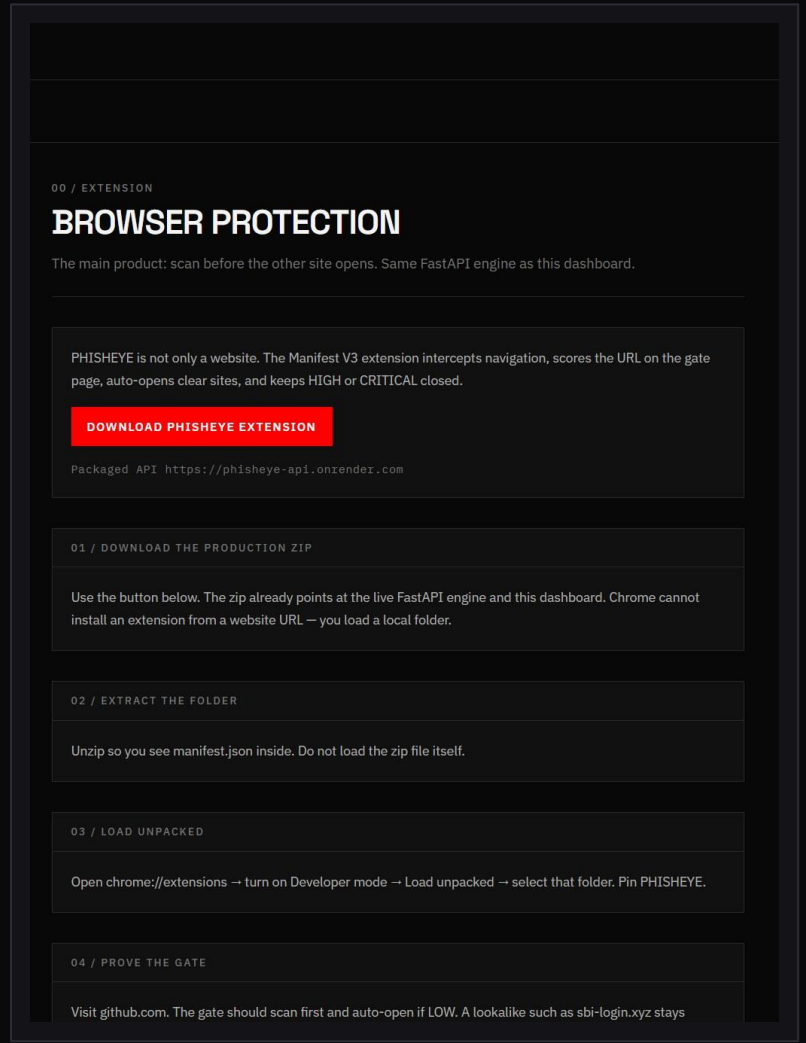
Extension

Visible links + navigation

Extension flow

User navigates
→ service worker holds tab
→ POST check-url
→ LOW: auto-open
→ MEDIUM: ask to continue
→ HIGH/CRITICAL: stay closed

MV3 1.8.0
Load unpacked — not Web Store
Does not read passwords or SMS



Two views. One score. Documented combination — not a secret formula.

SIMPLE VIEW

This website may be pretending to be a trusted service.
Do not enter your password, OTP, or card number.

Verdicts: NO MALICIOUS INDICATORS · BE CAREFUL · DANGER

TECHNICAL VIEW

Domain / lookalike tags · TLS · RDAP age
Threat-intel provider + match type
Heuristic reasons · psych findings
model_confidence = signal agreement

URL decision (floors)

Confirmed intel: max(heuristic, floor)
exact URL 85 · normalized 80
hostname 72 · registered domain 55
Heuristics cannot veto a confirmed hit.

Content fusion (weights)

url 0.22 · domain 0.16 · ml 0.10 · nlp 0.10
psych 0.10 · intel 0.10 · brand 0.08
redirect 0.04 · sender 0.05 · community 0.05
LOW 0–29 · MED 30–49 · HIGH 50–69 · CRIT 70–100

Names plus one job. No logo wall. No Postgres.

FRONTEND

React 19 dashboard
Vite 8 + Tailwind 4
Axios to FastAPI
Recharts on Statistics

BACKEND

FastAPI + Uvicorn
Pydantic request models
bcrypt passwords
JWT on mutating routes

DATA

Live: sqlite3 threats.db
users · url_checks · scans
Postgres = FUTURE, not live

INTELLIGENCE

Heuristics + fusion
Public threat feeds
Not a trained network
ml channel copies URL score

SECURITY

SSRF before unwrap
Parameterized SQL
5 MB upload cap
Extension does not read passwords

EXTENSION

Chrome MV3 1.8.0
Scan-before-open gate
Same check-url API
Load unpacked, not Web Store

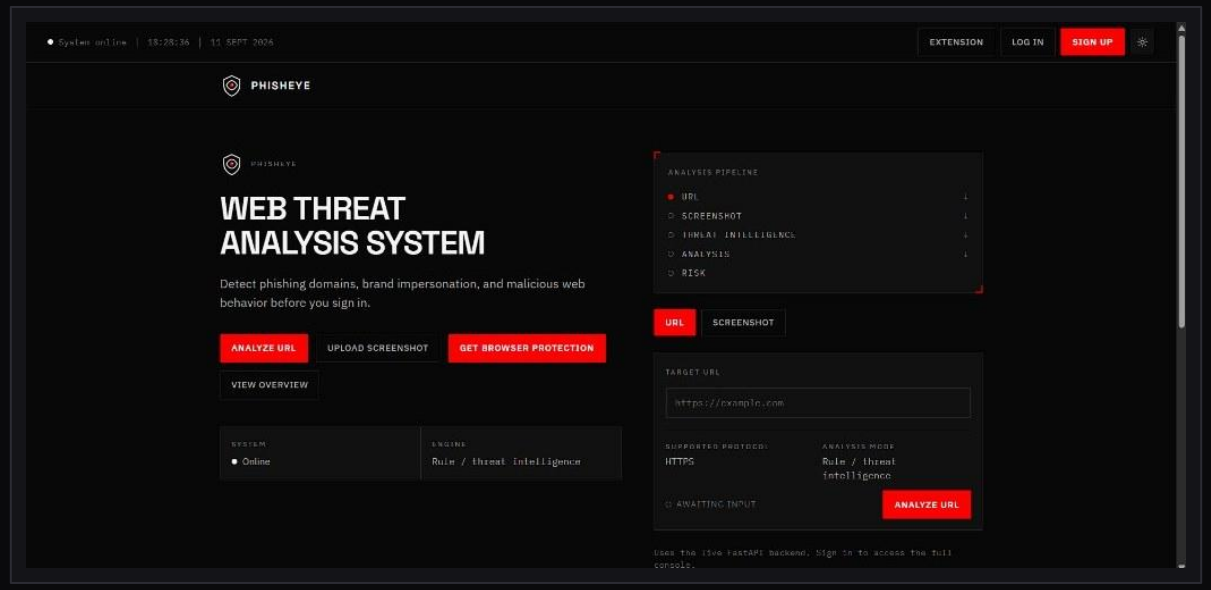
EXTERNAL

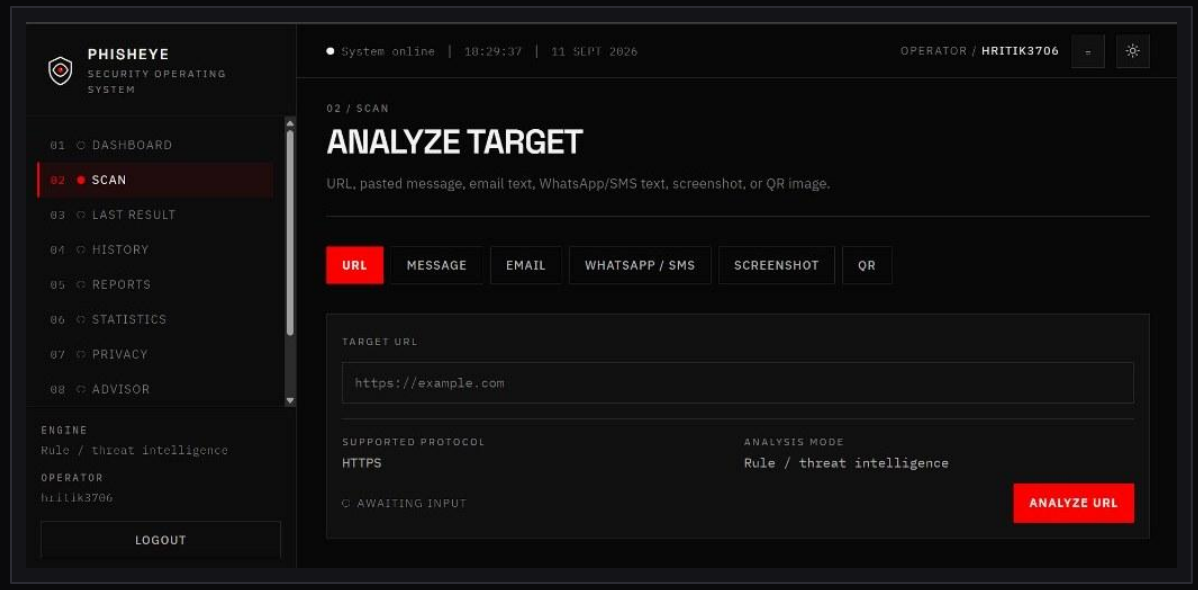
OpenPhish · URLhaus
PhishTank · Phishing Army
Optional Google Safe Browsing
Miss ≠ SAFE

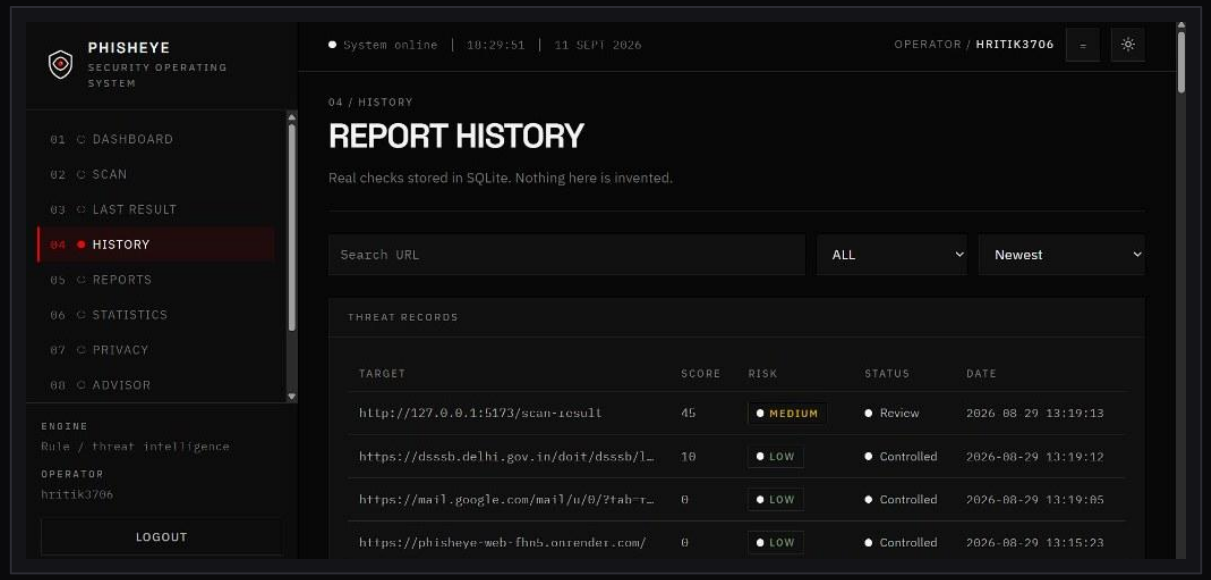
DEPLOY

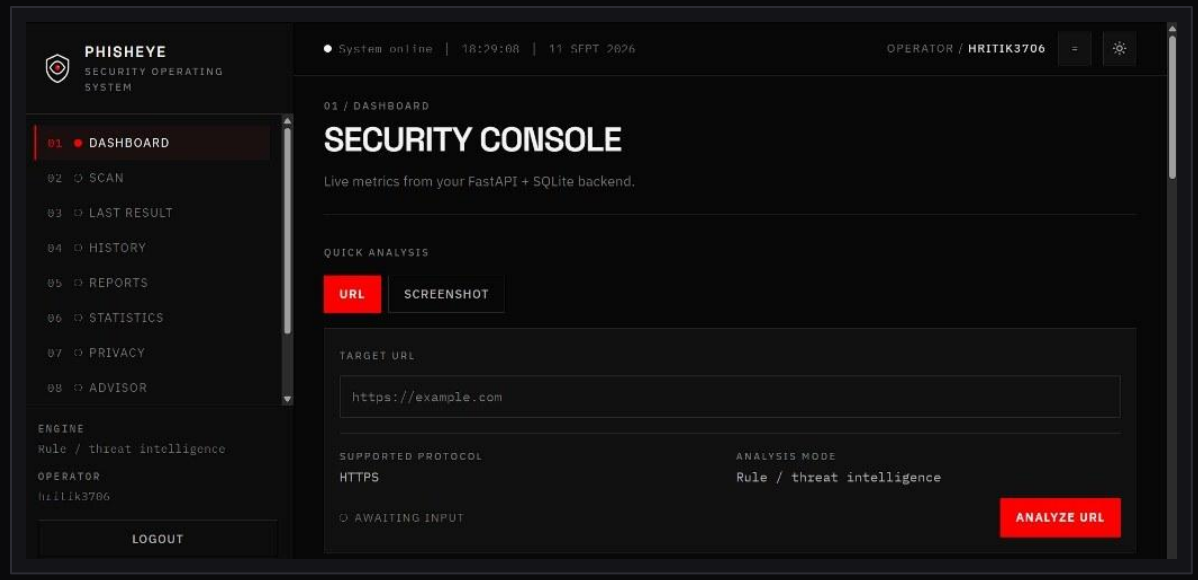
Render FastAPI service
Static React site
Extension zip on /extension
Free API sleeps ~15 min

Screenshots of the running UI. Dashboard counts are not an F1 score.









Protect people who cannot read a certificate warning.

Impact now (implemented)

Safer click path: scan before open
Plain-English OTP warnings
Multi-channel paste analysis
Community labels as a signal, not proof

FUTURE / NEXT PHASE

Curated reports → supervised model
PostgreSQL at scale
Android SMS companion
Official inbox APIs if granted

Not on this slide as current work.

Phishing + social engineering + threat intelligence + explainable scoring + browser protection.

That combination is PHISHEYE.